



SIGNFLOW

Project Documentation

**A modern digital document signing and management platform for individuals,
businesses, and teams**



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01. PROJECT AT A GLANCE

SignFlow is a digital document signing and management platform designed for individuals, businesses, and teams to upload, manage, send, sign, track, and complete documents digitally.

It brings document management, digital signatures, recipient tracking, signing workflows, audit trails, document activity, notifications, and analytics into one centralized workspace.

Core Modules

- Document Management
- Document Library
- PDF Preview
- Digital Signatures
- Signature Requests
- Recipient Management
- Signing Progress
- Document Tracking
- Audit Trail
- Notifications
- Document Verification
- Analytics
- Security Logs
- Search & Filters
- Settings

02. PROBLEM, USERS & PRODUCT VALUE

Problem

Traditional document signing often requires users to download documents, print them, sign them manually, scan them, and send them back.

For businesses and teams, managing multiple documents and recipients also makes it difficult to track pending signatures, completed agreements, document activity, and signing progress.

SignFlow brings the complete document signing workflow into one digital platform.

Target Users

Individual Users

Upload, sign, manage, and track personal documents.

Business Users

Send documents to recipients, manage signature requests, track progress, and maintain document records.

Teams

Collaborate around shared documents, monitor signing activity, and maintain document history.

Recipients

Receive signature requests, review documents, add signatures, and complete assigned signing actions.

Product Value

Centralized document management + Digital signing + Recipient tracking + Signing visibility
+ Audit history + Secure workflow

03. ARCHITECTURE & TECHNOLOGY

SignFlow uses a frontend-backend architecture where the frontend communicates with REST APIs connected to the application database and document services.

Frontend: Next.js + React

Backend: Node.js + Express.js

Database: MongoDB

ODM: Mongoose

Authentication: JWT Authentication

Password Security: bcrypt

File Uploads: Multer

Configuration: dotenv

API Security: CORS

Architecture

User → Next.js / React → REST API → Express.js → Mongoose → MongoDB

Core Backend Structure

- Controllers
- Middleware
- Models
- Routes
- Services
- Utilities
- Uploads
- Server Configuration

The frontend was first generated through Build and then taken into local development before backend and database functionality was implemented through CORA.

04. CORE E-SIGNATURE WORKFLOW

The main SignFlow workflow connects document upload, recipient management, digital signing, and completion tracking.

Authentication → Dashboard → Upload Document → Document Details → Invite Signers → Signature Request → Recipient Review → Digital Signing → Signing Progress → Completion → Audit Trail

Document Workflow

Users upload a document and create its document record.

Recipients are then invited to review and sign the document.

Signing Workflow

Document Uploaded → Signature Request Sent → Document Viewed → Document Signed → Workflow Completed

Users can:

- Create signatures
- Draw signatures
- Upload signatures
- Select saved signatures
- Place signatures
- Resize signatures
- Sign across multiple pages
- Invite multiple recipients
- Track recipient status
- Review the audit trail

The signing experience is centered around clear signature placement, recipient status, document progress, and activity tracking.

05. FEATURE MAP

Dashboard

Overview of total documents, pending signatures, completed documents, drafts, recent activity, and quick actions.

Document Management

Upload, organize, search, filter, preview, and manage documents.

Document Library

Central location for recent, shared, active, and completed documents.

PDF Preview

View documents with page thumbnails, zoom controls, annotations, version information, and document status.

Digital Signing

Create, draw, upload, place, resize, and apply signatures to documents.

Signature Requests

Send documents to recipients and monitor pending signature requests.

Recipients

Manage recipient information and signing status.

Signing Progress

Track document status from upload through completion.

Audit Trail

Maintain a chronological record of uploads, invitations, signatures, downloads, and completion events.

Notifications

Provide updates about signature requests, document activity, and pending actions.

Search

Search and filter documents based on relevant document information and status.

Document Verification

Provide document verification and security-related information.

Analytics

Display document turnaround time, completion rates, active requests, user activity, and workflow performance.

Settings

Manage profile, account, application, and configuration settings.

06. C0 WEB CHAT

C0 Web Chat was used for structured research, requirement analysis, product planning, and prompt refinement during the SignFlow workflow. It helped break down the document signing platform into clear product modules, user journeys, technical requirements, and implementation areas.

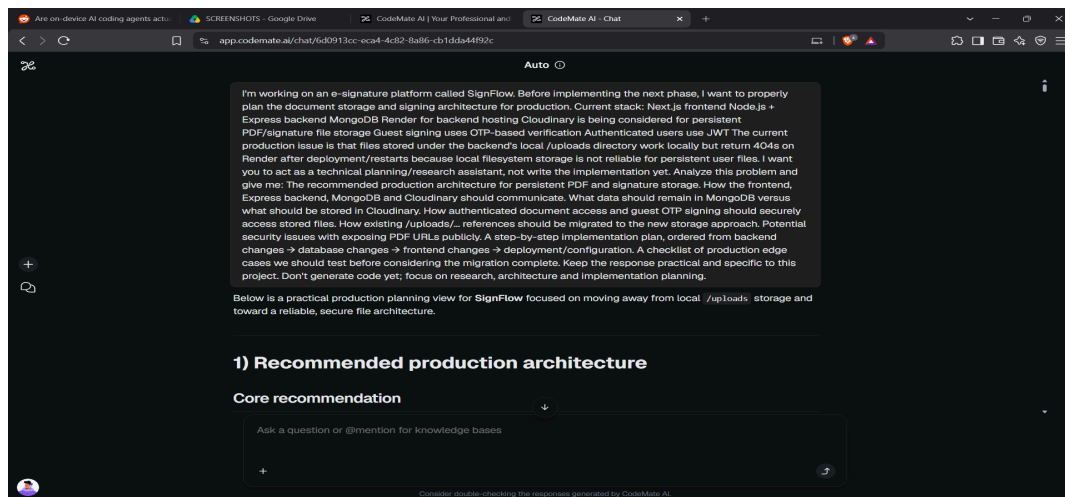
C0 Usage

- Product research
- Requirement analysis
- User-flow planning
- Feature planning
- Architecture analysis
- Workflow planning
- Documentation planning
- Prompt refinement
- Implementation planning

Workflow

Requirement → C0 Research & Analysis → Product Planning → Build → CORA Implementation → Review

C0 was used as the planning and analysis layer before moving the requirements into Build and CORA.



07. C0 PROMPT HIGHLIGHTS

C0 was mainly used for:

- Product research
- Requirement planning
- Feature breakdown
- Architecture analysis
- Document workflow planning
- Signing workflow analysis
- Technical planning
- Prompt refinement

Planning Areas

Foundation → Authentication → Documents → Recipients → Signing → Tracking → Audit → Analytics

Prompts Used

Prompt 1 : Product & Problem Research

Analyze SignFlow as a digital document signing and management platform for individuals, businesses, and teams. Break down the main problem being solved, target users, document signing challenges, recipient workflows, document management requirements, and the core value proposition. Identify how the platform can simplify the complete journey from uploading a document to sending signature requests, completing signatures, tracking progress, and maintaining an audit history.

Prompt 2 : Product Requirement Planning

Create a structured Product Requirements Document for SignFlow covering document management, document upload, PDF preview, digital signatures, signature requests, recipients, signing progress, audit trails, notifications, search, document verification, analytics, security logs, and settings. Organize the requirements around clear user workflows and explain how these modules should work together as one document signing platform.

Prompt 3 : Technical & Architecture Analysis

Analyze the technical requirements for building SignFlow as a production-ready document signing platform. Break down the frontend, backend, database, authentication, file upload system, document storage, REST APIs, recipient management, signing workflow, audit logging, and security requirements. Identify the dependencies between these components and create a clear technical structure that can be translated into implementation tasks.

Prompt 4 : Workflow & Implementation Planning

Review the SignFlow requirements and create an incremental implementation plan for the complete document signing workflow. Prioritize authentication and document management first, followed by document upload, recipient management, digital signing, signing progress, audit trails, notifications, analytics, and document verification. Identify dependencies between modules and organize the implementation so each stage can be tested before moving to the next.

08. BUILD: DESIGN MODE + PROTOTYPE MODE

Design Mode

Build Design Mode was used to establish the visual direction and complete product experience for SignFlow.

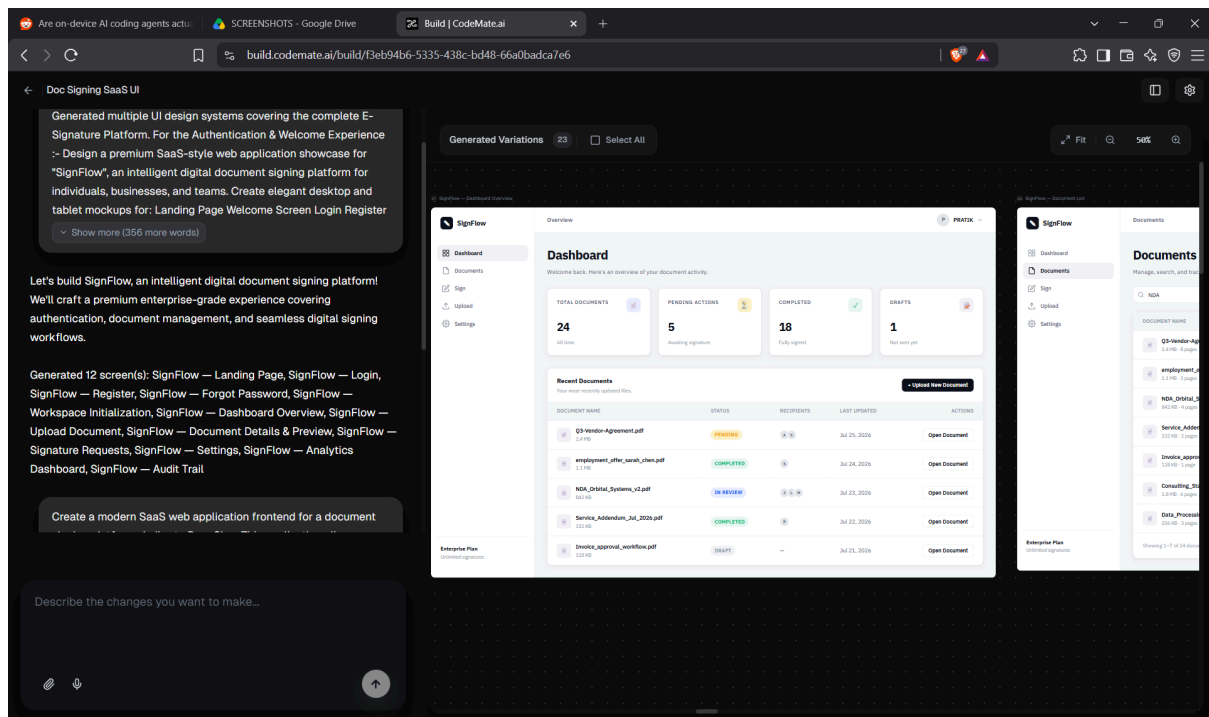
The design focused on:

- Premium SaaS interface
- Enterprise visual language
- Clean professional layouts
- Blue, white and emerald color palette
- Professional typography
- Consistent spacing
- Responsive layouts
- Reusable UI components
- Document-focused workflows
- Digital signing interactions
- Productivity-oriented workspace

Design Experience

The design covered:

- Landing Page
- Login
- Register
- Forgot Password
- Email Verification
- Account Setup
- Workspace Initialization
- Dashboard
- Document Library
- Document Upload
- PDF Preview
- Document Details
- Signature Requests
- Signature Canvas
- Signature Library
- Recipient Management
- Signing Progress
- Audit Trail
- Security Logs



Prototype Mode

The approved SignFlow design was then taken forward into application development.

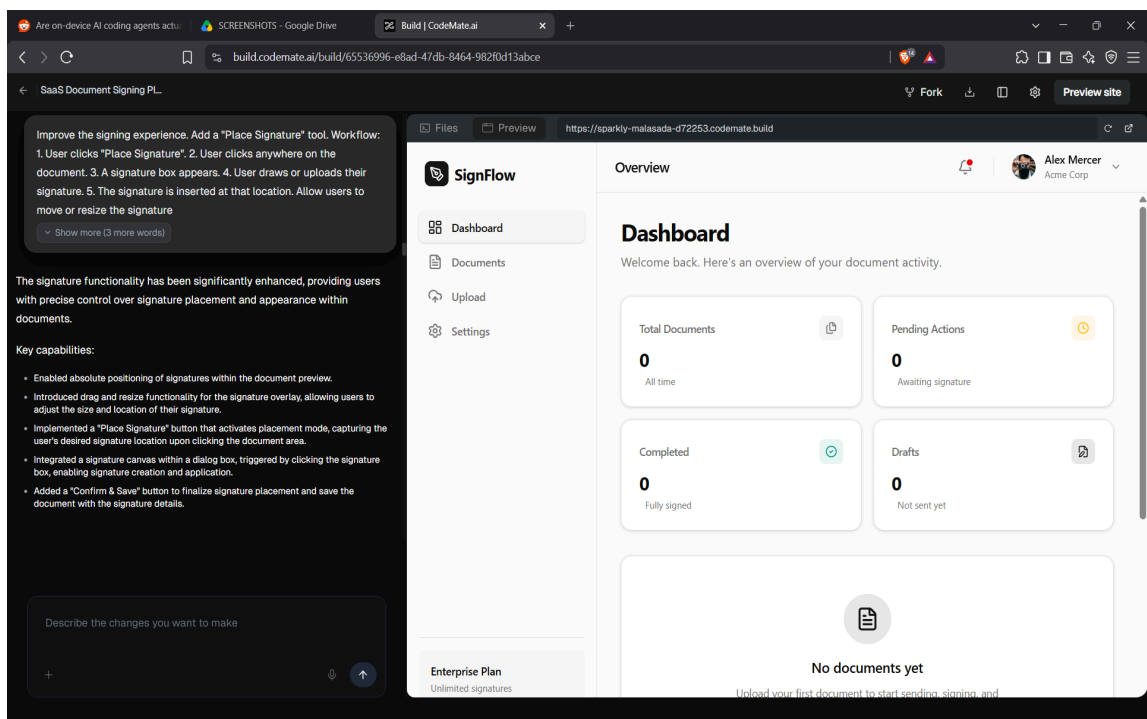
The frontend implementation focused on converting the approved interface into a functional document signing application using reusable components and mock data before backend integration.

Implementation

- Dashboard
- Document List
- Upload Document
- Document Viewer
- Sign Document
- Document Tracking
- Settings
- Search
- Filters
- Signature Canvas
- Signature Placement
- Status Tracking
- Reusable Components
- Responsive Layouts

Reusable Components

- Sidebar
- Navbar
- Dashboard Cards
- Document Table
- Upload Component
- Signature Canvas
- Status Badge
- Signature Placement Modal
- Forms
- Buttons



09. BUILD PROMPT HIGHLIGHTS

The Build prompts focused on creating a complete digital document signing platform rather than a generic SaaS dashboard.

Main Prompt Focus

- Digital document signing
- Document management
- Signature workflows
- Recipient tracking
- PDF preview
- Document status
- Audit history
- Responsive SaaS UI
- Reusable components
- Mock data

Prompts Used

Prompt 1 : Design Foundation

Design a premium SaaS-style web application called SignFlow, an intelligent digital document signing platform for individuals, businesses, and teams. Create a professional experience covering the landing page, authentication flow, account setup, workspace initialization, document management, digital signing, recipient workflows, audit trails, analytics, and security. Use a sophisticated blue, white, and emerald visual system with strong typography, consistent spacing, responsive layouts, and reusable components.

Prompt 2 : Document Management Workspace

Design a modern enterprise workspace for users to manage and sign digital documents. Include Dashboard, Upload Document, Drag & Drop Upload, Recent Documents, Document Library, PDF Preview, Document Details, Signature Requests, Pending Actions, Completed Documents, Shared Documents, Notifications, Activity Feed, Search Documents, Profile, and Settings. Focus on document summaries, pending signatures, completed agreements, recent uploads, storage usage, quick actions, document previews, annotations, version history, and collaboration status.

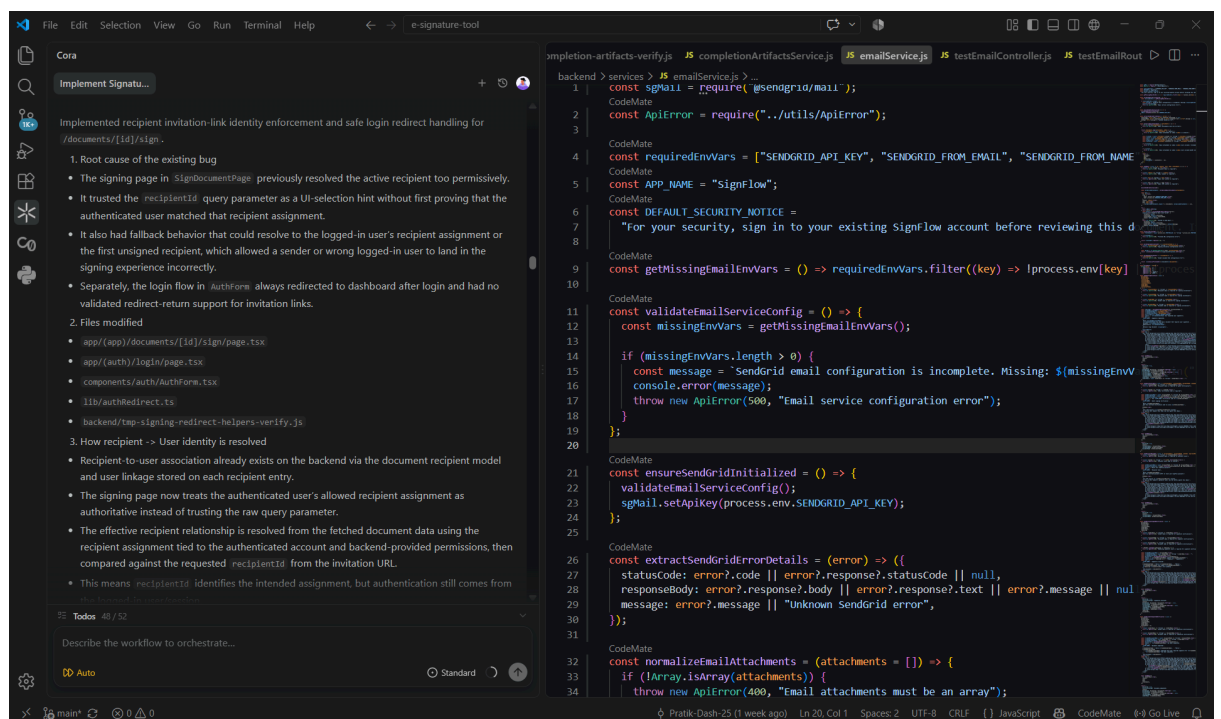
Prompt 3 : Digital Signing Experience

Design a complete digital signing experience for document owners and recipients. Include Signature Canvas, Create Signature, Upload Signature, Signature Library, Place Signature, Drag & Resize Signature, Multi-page Signing, Invite Signers, Signing Progress, Recipient Status, Audit Trail, Signing

Timeline, Document Verification, Security Logs, and Analytics Dashboard. Make signature placement intuitive and ensure users can clearly understand recipient status and document progress throughout the signing workflow.

Prompt 4 : Functional Frontend

The SignFlow visual design has been finalized. Convert the approved design into a functional responsive frontend without changing the established visual system. Implement authentication, dashboard, document management, upload document, PDF preview, digital signing, document tracking, settings, search, filters, signature placement, status tracking, reusable components, and responsive layouts. Use mock JSON data initially so the complete frontend workflow can be verified before backend API integration.



10. CORA: PLANNING & IMPLEMENTATION

CORA was used as the main development and implementation layer after the Build frontend was generated.

The frontend source was taken into local development and CORA was then used to progressively implement the backend, database, authentication, APIs, document management, recipient workflows, and live document operations.

CORA was used for:

- Backend setup
- Database models
- MongoDB integration
- Authentication
- REST APIs
- Document management
- File uploads
- Document persistence
- Recipient tracking
- Audit logging
- Frontend API integration
- Replacing mock data
- End-to-end document workflow

Workflow

Build Frontend → CORA Planning → Backend Setup → Database → Authentication → API Integration → Live Workflow → Testing

Prompts Used

Prompt 1 : Backend Foundation

Analyze the existing SignFlow frontend and generate a production-ready backend using Node.js, Express.js, MongoDB with Mongoose, JWT authentication, bcrypt, dotenv, multer, and CORS. Create a clean backend structure containing configuration, controllers, middleware, models, routes, services, utilities, uploads, server configuration, and package dependencies. Keep the existing frontend unchanged and create a scalable foundation for authentication, document management, file uploads, recipient workflows, audit logging, and future API integration.

Prompt 2 : Database & Document Models

Generate the MongoDB models required for the SignFlow document signing platform using Mongoose best practices. Create User, Document, Recipient, and AuditLog models with the required fields and relationships. Ensure documents are connected to their owners and recipients, recipient status and signature information can be persisted, and every important document event can be recorded through audit logs. Keep the schema structure modular and suitable for the existing backend architecture.

Prompt 3 : Authentication Implementation

Implement the SignFlow authentication module with registration, login, and protected user profile access. Use JWT authentication and bcrypt password hashing with authentication middleware, request validation, centralized error handling, and protected API routes. Create the required controllers, routes, middleware, and services while keeping authentication modular and secure. Ensure authenticated users can access protected document management functionality.

Prompt 4 : Replace Mock Data with Live APIs

Implement the document workflow using the existing backend APIs. Replace the current mock upload and document creation logic with real API calls. Upload the PDF through the backend upload endpoint, use the upload response to create the document record, display backend validation errors, and redirect to the Documents page after successful creation. Replace mock Dashboard and Documents data with backend responses while keeping the existing UI unchanged. Verify the complete upload, creation, retrieval, and listing workflow after integration.

11. CORA TASK ARCHIVE

Instead of listing every individual CORA task, the implementation can be grouped into the major areas below.

Backend

- Node.js backend
- Express.js server
- Controllers
- Routes
- Middleware
- Services
- Utilities
- File upload handling
- Environment configuration
- CORS

Database

- MongoDB
- Mongoose
- User model
- Document model
- Recipient model
- AuditLog model
- Document relationships
- Persistent document records

Authentication

- Registration
- Login
- JWT
- Password hashing
- Authentication middleware
- Protected routes
- User profile
- Request validation
- Error handling

Document Workflow

- PDF upload
- Document creation
- Document storage
- Document retrieval
- Document listing
- Dashboard data
- Document details
- Document tracking
- Live API data

Recipient Workflow

- Recipient information
- Signature requests
- Recipient status
- Signing progress
- Signature completion

Audit & Activity

- Document events
- Upload history
- Signature activity
- Recipient activity
- Audit records
- Timestamps
- Metadata

Frontend Integration

Mock document data was replaced with live backend APIs while keeping the existing frontend architecture and approved UI intact.

12. IMPLEMENTATION MILESTONES

Phase 1: Research

Used C0 Web Chat for product research, requirement analysis, workflow planning, and technical direction.

Phase 2: Design

Created the initial SignFlow UI system using Build Design Mode.

Phase 3: Frontend

Converted the approved design into a functional frontend with authentication, dashboard, document management, signing, tracking, and settings screens.

Phase 4: Backend

Created the Node.js and Express.js backend architecture with controllers, routes, middleware, services, utilities, and file upload handling.

Phase 5: Database

Created MongoDB models for users, documents, recipients, and audit logs.

Phase 6: Authentication

Implemented registration, login, JWT authentication, password hashing, protected routes, validation, and error handling.

Phase 7: API Integration

Connected the frontend to the backend and replaced mock document workflows with live API operations.

Phase 8: Document Workflow

Implemented document upload, storage, retrieval, document creation, document tracking, and persistent document data.

Phase 9: Verification

Verified the complete document workflow from upload through document retrieval and signing-related operations.

13. BACKEND, DATA & INTEGRATIONS

SignFlow uses MongoDB with Mongoose to persist users, documents, recipients, and document activity.

Core Data

User

- Name
- Email
- Password
- Role
- Created At

Document

- Title
- Owner
- File Name
- File Path
- Status
- Recipients
- Created At

Recipient

- Name
- Email
- Status
- Signed At
- Signature Image

Audit Log

- Document ID
- Action
- Performed By
- Timestamp
- Metadata

Backend Flow

Frontend → REST API → Express.js → Controller → Service → Mongoose → MongoDB

Document Flow

Upload PDF → Backend Upload Endpoint → Document Creation → MongoDB → Document Retrieval → Frontend

The application moved from mock document data to backend-driven document upload, creation, retrieval, and persistent storage.

14. AI ASSISTANT & WORKSPACE CONTEXT

The SignFlow documentation does not identify a separate AI Assistant as a core product module.

The primary intelligent workflow is centered around digital document processing, signing, recipient management, tracking, verification, and audit history.

Product Context

- Document
- Recipients
- Signature Request
- Signing Activity
- Completion
- Audit Trail

Workspace Context

The authenticated user operates within the application workspace where documents, recipients, signing activity, and document history are managed.

Access to document-related functionality is protected through the authentication layer.

15. RESPONSIVE UI & POLISH

SignFlow was designed as a responsive enterprise SaaS application with a consistent visual system across document management and digital signing workflows.

UI Characteristics

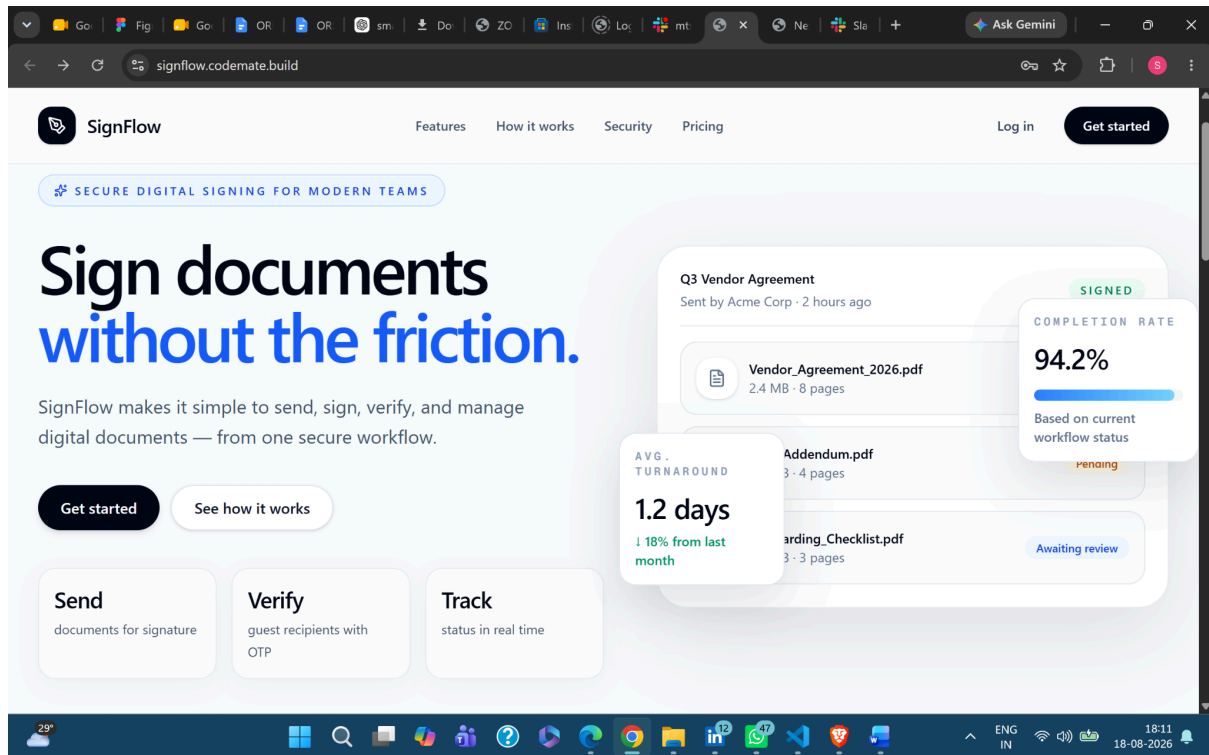
- Responsive layouts
- Desktop and tablet support
- Premium SaaS styling
- Professional typography
- Consistent spacing
- Rounded components
- Soft shadows
- Blue, white and emerald design direction
- Reusable components
- Smooth interactions
- Clear status indicators
- Document-focused layouts
- Signing-focused interactions

Responsive Experience

Desktop → Tablet → Mobile

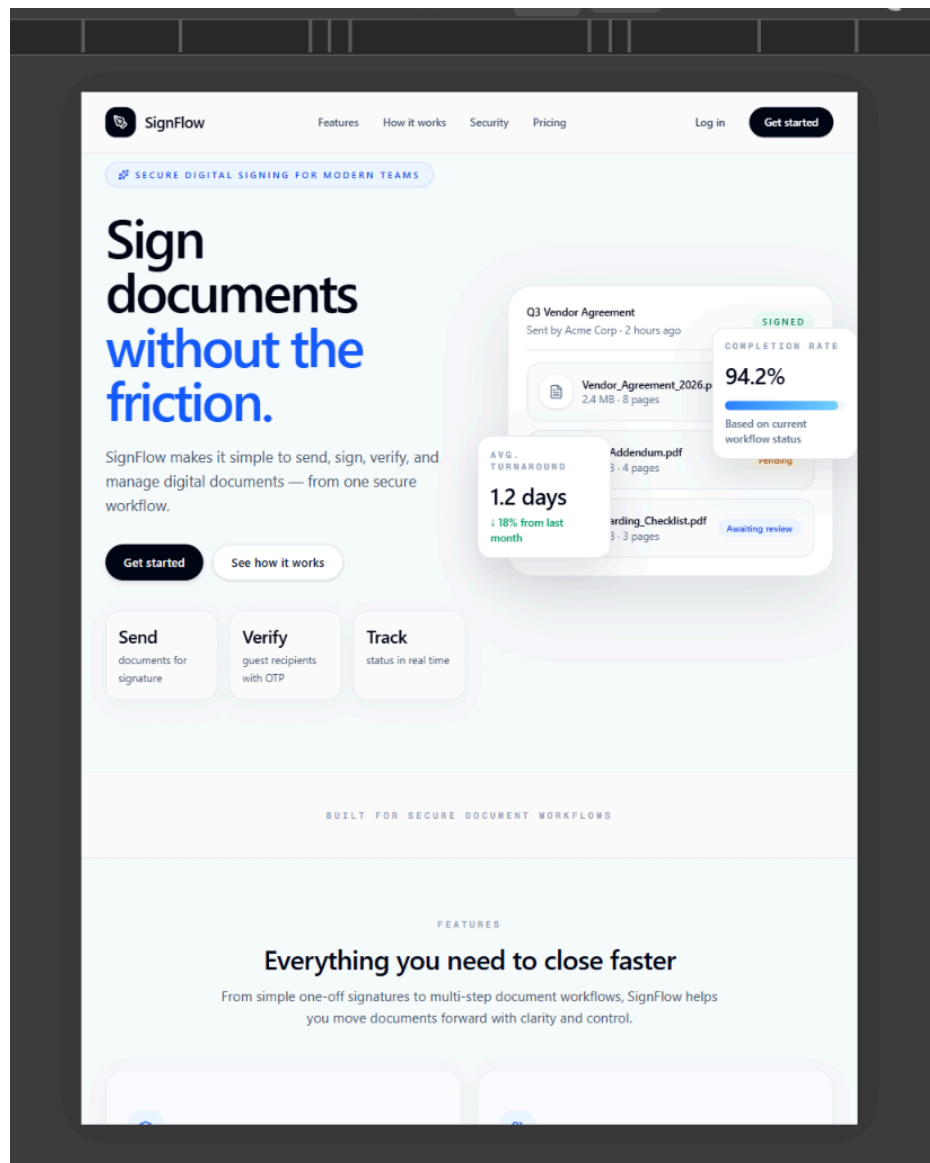
Desktop

Full document workspace with sidebar navigation, dashboards, document tables, PDF previews, signing interfaces, and multi-column layouts.



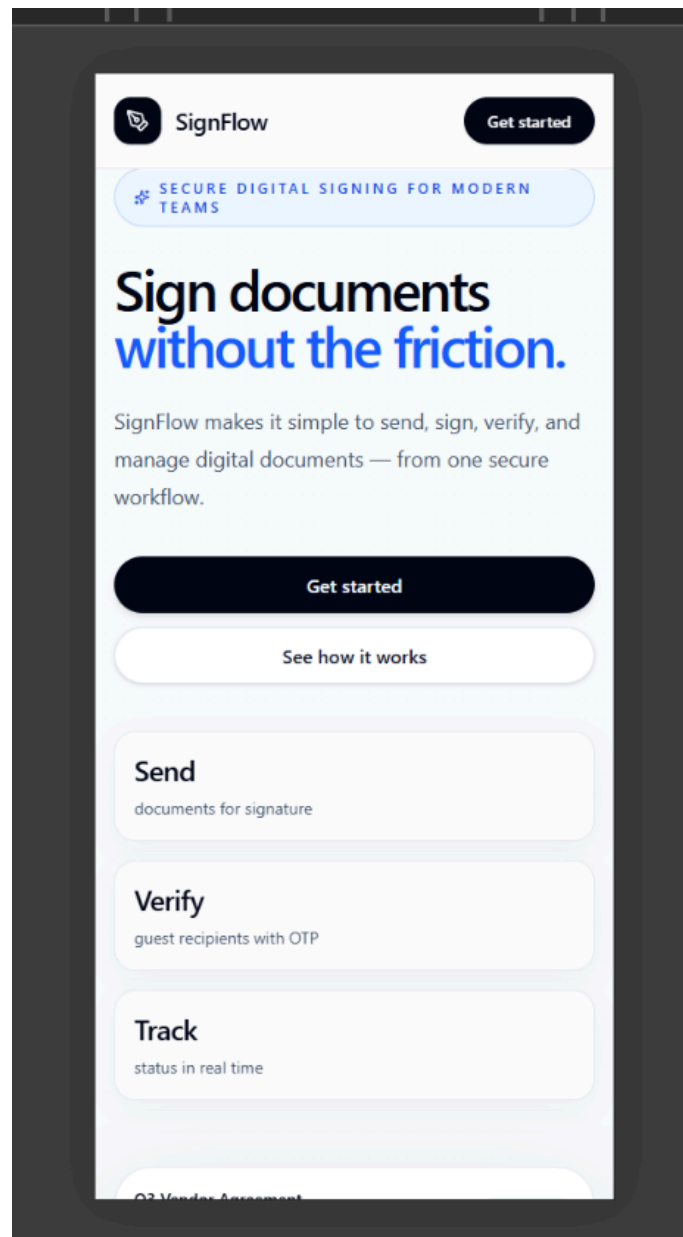
Tablet

Adjusted navigation and content density while preserving the main document workflow.



Mobile

Stacked layouts, simplified navigation, and touch-friendly document and signing interactions.



Final Product Flow

C0 Web Chat → Build Design Mode → Build Code / Prototyping → CORA Development → Final Product Preview
